

Research Experience

Indian Institute of Management Kashipur (IIM Kashipur)

- May, 2025 –
Aug, 2025
- Crop Residue-to-Industry Recommendation Model.**
Developed a recommendation model for predicting optimal industrial utilization pathways for agricultural crop residues using data-driven techniques, aiming to improve sustainable biomass management and resource utilization.
- 2025
- Live Project: Supply Chain Optimization Tool.**
 - Developed an optimization-based decision support tool for a supply chain company with 400+ employees.
 - Designed scheduling and workforce coordination system to improve task allocation efficiency.
 - Implemented route optimization for field employees to minimize travel time and operational costs.
 - Utilized Gurobi optimization solver for mathematical modeling and cost minimization.
 - Improved overall operational efficiency through data-driven resource planning and scheduling.
- Advisor :
- Prof. Sunil Kumar Jauhar,** *Indian Institute of Management Kashipur*

Indian Institute of Science Education and Research (IISER) Kolkata

- 2024
- Spectral Body Wave Analysis for Source Parameters Estimation.**
Conducted spectral analysis of body seismic waves using **SAC (Seismic Analysis Code)** to estimate earthquake source parameters. Performed signal processing and spectral modeling to analyze seismic wave characteristics for earthquake studies.
- Advisor :
- Prof. Supriyo Mitra,** *Department of Earth Sciences, IISER Kolkata*
- 2024
- Seismic Data Analysis Calculator.**
Developed computational tools for seismic data analysis, including epicentral distance calculation, earthquake location estimation, and seismic hazard assessment. Designed data-driven workflows to improve earthquake analysis and disaster preparedness.
- 2023
- Sedimentology Fieldwork, Himalayan Siwalik Basin.**
Applied quantitative data analysis techniques to sedimentary rock samples collected from the Himalayan Siwalik Basin to investigate depositional history, sedimentary processes, and basin evolution, providing insights into past environmental changes.
- Advisor :
- Prof. S. K. M. Mandal,** *Department of Earth Sciences, IISER Kolkata*

Relevant Coursework

Probability and Statistics	Linear Algebra
Data Structures and Algorithms	Calculus and Optimization
AI / ML	Database Management and Warehousing

Academic Achievements & Recognitions

- 2024
- International Youth Math Challenge 2024 (IYMC 2024)** (Germany), completed both the Qualification Round and Pre-Final Round.

2021 **Qualified JEE Main**, ranked in the top 0.1% of candidates.

2021 **Qualified IISER Aptitude Test (IAT)**, ranked in the top 0.06% of candidates.

Computer skills

Programming

Languages Python, PyTorch, R, C, C++,

Web

Technologies HTML 5, PHP, JSP, Javascript

Database SQL, MySQL, Apache

Tools Gurobi, Git, LATEX, Numpy/Scipy, Pytorch,

Position of Responsibility

2023 **Event Management & Logistics**, *cult fest -IISER Kolkata.*

2022 **Organizer, Team Member: E-Cell**, *IISER Kolkata.*

◦ Sponsored Team (Inquivesta - Sci Fest)

2022 **Hostel Representative**, *IISER Kolkata.*

Education

2021 – 2025 **BS in Earth Science**, *Indian Institute of Science Education and Research (IISER) Kolkata,*
CGPA: 6.7

Interdisciplinary science - climate data science & analytics

2020 **Class 12 : 83.8%**, *College of Commerce, Arts & Science, Patna.*

2018 **Class 10 : 82.8%**, *Himalayan Residential School, Patna.*

Referees

Dr. Sunil Kumar Jauhar

Assistant Professor,

Operations Management & Decision Sciences

IIM KASHIPUR

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